

Chromato-Textural Profiling for Unroasted Robusta Coffee Bean Grading: An Interpretable Framework via Chaos Game Optimization and XGBoost

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Abstract. This work addresses the assessment of coffee bean quality, which is a major determinant of market value. Admittedly, traditional manual grading is a labor-intensive, subjective process that is prone to inconsistencies. In this respect, we propose a robust and interpretable computer-vision framework for grading Robusta coffee beans in the context of precision agriculture. Using a purposefully curated dataset containing 3,859 high-resolution images from the Coorg region of India, a two-stage processing pipeline was developed: First, a YOLOv11m-seg model was used solely for ROI extraction by attaining a segmentation mean Average Precision (mAP@50) of 0.995, thus ensuring the precise elimination of beans from the background noise. Then, the extraction of an extensive 48-dimensional feature vector was performed that characterized the CIELAB color statistics along with textural descriptors of LBP, GLCM, Gabor, and morphological shape characteristics. To eliminate redundant features without jeopardizing interpretability and, at the same time enhance the computational efficiency of our pipeline, a hybrid classifier architecture has been proposed which integrates Chaos Game Optimization (CGO) using Optuna with an XGBoost classifier. The metaheuristic optimization offered by CGO efficiently reduced the feature space into a compact subset of 30 discriminative features. Empirical results indicate the superiority of the proposed CGO–XGBoost framework in classification with a testing accuracy of 93.61% and an F1-score of 0.9359, outperforming the baseline ensemble methods and conventional evolutionary optimization strategies such as Genetic Algorithms. These empirical findings thus indicate that the proposed methodology offers a scalable low-latency, explainable solution that can easily be deployed on edge computing devices in agricultural fields and industrial sorting lines.

Keywords: Coffee Bean Grading, Computer Vision, Chaos Game Optimization (CGO), XGBoost, Precision Agriculture, Feature Selection, YOLOv11.

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1 Introduction

The Strict quality-control procedures that support price, market value, and customer satisfaction are the foundation of the global coffee business. The grading of unroasted or green coffee beans is one of the most important processing steps in achieving the best possible cup quality. Conventional assessment is carried out visually by trained experts. However, as identified by researchers across the field, a human-centered approach tends to be subjective, labor-intensive, time-consuming, and prone to fatigue and error [1,6,12]. The demand for objective, consistent, and fast grading mechanisms has driven the shifting interest in automated computer-vision and ML systems.

So far, the development of automated coffee-grading technology has passed through two paradigms: classical machine learning based on handcrafted features and, lately, deep learning (DL) architectures. Early work made a case for imaging-based methods by initially considering only basic visual appearance. In a seminal work, Turi et al. [1] showed that color, morphology, and texture together could be effectively applied to the geographical origin classification of Ethiopian coffee beans. This basis was extended by de Oliveira et al. [3], who used the CIELAB color space and Bayesian classifiers to reach high precision in color-based quality sorting. Since color information alone cannot detect complex physical defects, textural analysis started to be adopted in later works as well. Montes Condori et al. [2] presented the Color Gray Level Co-occurrence Matrix (CGLCM) with Support Vector Machines (SVM), whereas Akbar et al. [7], on the other hand, incorporated explicit use of Local Binary Patterns (LBP) for the capture of micro-textures in Arabica beans.

These traditional techniques offered interpretability, but they frequently lacked sophisticated optimization and feature representation. For instance, systems created by García et al. [6] with KNN were able to be implemented practically in embedded hardware, but they relied on basic geometric features rather than sophisticated frequency-domain descriptors like Gabor or Wavelet transforms [4, 6]. This paradigm was drastically altered with the advent of deep learning. Pinto et al.'s use of CNNs for defect classification is among the first instances of DL's ability to automatically learn hierarchical representations [5]. The trend accelerated: Chang and Huang [8] modified AlexNet architectures for texture capture, while Arwatchananukul et al. [10] used MobileNetV3 for extensive defect categorization. More recently, Jiao et al. [12] established state-of-the-art performance by utilizing the Swin Transformers, Swin-HSSAM, that attain remarkable accuracy due to multi-scale feature fusion. In parallel, frameworks like YOLO have been applied for real-time sorting. For instance, Luis et al. [9] present a YOLO-based system that achieves an accuracy of 95.11%, whereas Ji et al. [11] explore enhanced YOLOv8 models for efficient detection of defects. Despite substantial advances, there still exist considerable knowledge gaps, particularly regarding the trade-off between computational efficiency and classification accuracy for edge deployment. Deep learning models give good accuracy but require considerable computation and large volumes of annotated datasets, reducing their viability in resource-poor regions of coffee production. The more classical machine learning methods are computationally efficient but have received lower results in the past due to suboptimal feature selection and insufficiency in the use of modern ensemble mod-

els such as XGBoost. What is more, even though chaotic metaheuristics showed a very great promise for other agriculture-related tasks, algorithms such as Chaos Game Optimization have not been explored so far for coffee-bean feature selection. Most important, the majority of studies deal with the Arabica variety, leaving a significant gap with respect to automatic grading solutions intended for beans belonging to Robusta variety with more complex surface textures.

Proposed Contribution. In view of the above-mentioned shortcomings, the present research proposes a hybrid model that combines deep learning-based localization with interpretable, optimized machine learning classification. This ensures precise segmentation, carried out by a YOLOv11m-seg module, whereas the classification relies on a mathematically sound feature space further refined with chaotic optimization. The significant contributions of this work are three-fold:

1. **Two-Stage Pipeline:** This is a modular architecture which separates bean localization, YOLOv11m-seg, from classification to provide clean, noise-free ROIs to the downstream analysis.
2. **Chromato-Textural Profiling:** A domain-specific feature vector of 48 dimensions containing all the chromatic and structural variability of the Robusta beans by using Lab color statistics, LBP, GLCM, Gabor filters, Wavelet transforms, and Fourier descriptors.
3. **Hybrid Optimization Strategy:** Novel integration of Chaos-Generating Optimization with the Optuna-tuned XGBoost classifier produced a compact feature subset of 30, outperforming the standard Genetic Algorithms in both accuracy and stability.

The rest of the paper is organized as follows: in Section 2, the dataset and the CGO-XGBoost methodology are described; Section 3 provides experimental results and ablation studies; conclusions, limitations, and future work are drawn out in Section 4.

2 Materials and Methods

2.1 Dataset Acquisition and Curation

This work uses the custom dataset authored by us “An Image Dataset of Pre-Roast Robusta Coffee Beans with Polygon Annotations for Automated Grading” (Hugging Face DOI: 10.57967/hf/6955) [13], consisting of 3,859 high-resolution single-bean images sourced from plantations in Coorg, Karnataka, India. Each bean was divided into four commercial categories by a panel of skilled local graders: A (Premium), B (Good), C (Standard), and D (Defective).

A constructed lightbox with diffused LED lights and a neutral gray/white background was utilized to record images in a controlled setting. Acquisition covered a variety of smartphone cameras (Samsung S23/S24, iPhone 15 Pro, Google Pixel 7, Poco X3 Pro) in order to confirm hardware robustness.

Data are grouped into four grade-specific directories, each holding exactly one bean per image. A selection of 2,192 photos includes high-precision polygon masks created with LabelMe. Table 1 illustrates the distribution of photos and annotations.

Table 1. Distribution of annotated images across grades

	A	B	C	D	Total
Category	Premium	Good	Standard	Defective	—
Total Images	1000	1000	1000	859	3859
JSON Masks	600	600	487	505	2192

2.2 ROI Segmentation and Bean Extraction

The classifier pipeline works on isolated region-of-interest crops, rather than whole-scene images; a robust segmentation stage is hence required. To generate consistent, background-free inputs, a deep learning-based approach was adopted.

Single-Class Annotation Strategy. Although grade labels are available, segmentation is treated as a class-agnostic problem. All annotated labels, such as grade_a and grade_b, are merged into a single “coffee” class. This approach prevents grade leakage during localization and maximizes the effective training pool.

Bean-Finder Workflow. A YOLOv11m-seg model—herein referred to as “Bean-Finder”—was trained on 2,192 polygon-annotated images based on an 80:20 train-validation split and a 640×640 input resolution, with the default loss configurations of Jocher et al. [14]. Then, the obtained model was applied on the entire dataset of 3,859 images. As shown in Fig. 1, the pipeline outputs a segmentation mask and extracts a tight ROI crop for every bean.

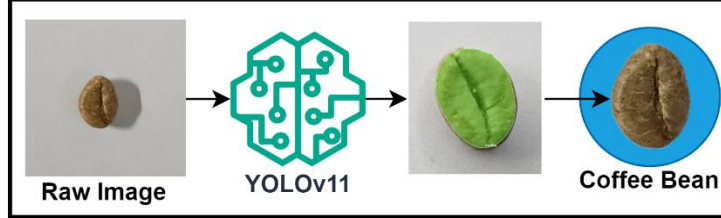


Fig. 1. Illustration of raw image, segmentation mask, and ROI crop.

2.3 Feature Extraction Methodology

It is necessary to quantify small visual cues that are not consistently separable in raw picture space in order to accurately grade coffee beans (CGA, CGB, CGC, and CGD). Each segmented bean image is converted into a compact, discriminative 48-dimensional representation via our feature extraction technique. The pipeline leverages information from four complementary domains (Fig. 2): color statistics, texture, multi-resolution frequency content, and morphological shape.

Color Statistics (CIELAB Space). As RGB is highly correlated and not perceptually uniform, we transform all images to the CIELAB (Lab) color space. For each channel, we calculate the mean(μ) and standard deviation(σ). This gives a 6-dimensional color vector which isolates color defects.

Texture and Surface Analysis. The texture features were extracted in order to capture surface roughness and grain microstructure.

- **Local Binary Patterns (LBP):** The uniform LBP ($P=8$, $R=1$) provides a 10-dimensional histogram encoding the surface roughness.
- **Gray-Level Co-occurrence Matrix (GLCM):** Computed at distance $d = 1$ and angle $\theta = 0^\circ$. From the normalized matrix, six statistics were extracted: contrast, dissimilarity, homogeneity, energy, correlation, and ASM.
- **Gabor Filters:** Images were convolved with Gabor kernels at four orientations: $\theta = 0, \pi/4, \pi/2, 3\pi/4$. The mean and variance were computed for each orientation, yielding an 8-D descriptor sensitive to fractures.

Multi-Resolution Frequency Analysis. DWT with the Haar basis was performed. The number of decomposition levels chosen was one, so four sub-bands were produced, namely LL, LH, HL, HH. The mean and standard deviation were calculated in each sub-band. By concatenating these features, an 8-dimensional feature set was obtained.

Morphological Shape Descriptors. For each bean, the biggest contour was extracted. The Fourier Descriptors (FD) were computed by retaining the magnitudes of the first 10 low-frequency harmonics, resulting in a 10-dim signature that captures broken or misshapen beans.

Feature Fusion. All domain-specific vectors were concatenated into a unified 48-dimensional representation as represented in the Eq. (1).

$$F_{\text{ALL}} = F_{\text{Color}} \oplus F_{\text{LBP}} \oplus F_{\text{GLCM}} \oplus F_{\text{Gabor}} \oplus F_{\text{Wavelet}} \oplus F_{\text{Fourier}} \quad (1)$$

This fusion integrates global and local information, enabling traditional machine learning models to exploit complementary cues from color, texture, multi-scale structure, and geometry. A summary of the extracted feature vectors is provided in Table 2

Table 2. Summary of Extracted Feature Vectors

Feature Domain	Algorithm/ Method	Dimensionality	Description
Color	CIELAB Statistics	6	Mean (μ) and Std Dev (σ) for L, a, b channels.
	Uniform LBP	10	Normalized histogram of uniform LBP patterns ($P=8$, $R=1$).
Texture	GLCM	6	Contrast, Dissimilarity, Homogeneity, Energy, Correlation, ASM.
	Gabor Filters	8	Mean and Variance across 4 filter orientations (θ).
Frequency	DWT (Haar)	8	Mean and Std Dev from Approx, Horizontal, Vertical, Diagonal sub-bands.
Shape	Fourier Descriptors	10	Magnitude of first 10 low-frequency contour harmonics.
Total	Fused Vector (ALL)	48	Total handcrafted features per sample.

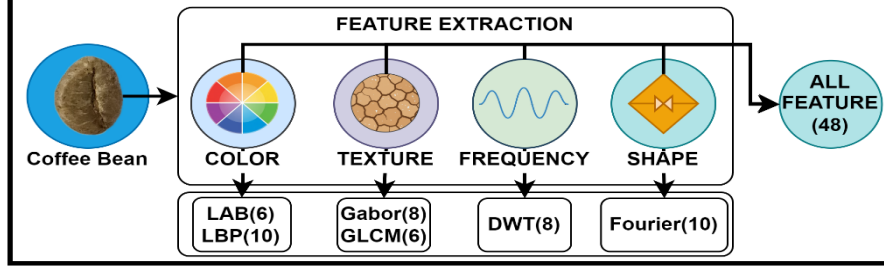


Fig. 2. Feature extraction framework combining color, texture, frequency, and shape descriptors into a unified 48-dimensional representation.

2.4 Proposed Methodology: CGO-Optimized XGBoost Framework

The aim of the paper is to handle the challenges imposed on precision agriculture by high-dimensionality feature redundancy and the strict low-latency classification requirement using a hybrid optimization framework. It proposes the combination of Chaos Game Optimization for discriminative feature selection with Optuna for hyperparameter tuning, which thus leads to a robust XGBoost classification model.

Section 3 gives the empirical validations of the proposed architecture, where the CGO-XGBoost framework attained a testing accuracy of 93.61% and F1-score of 0.9359, outperforming standard evolutionary approaches like GA and AOA, and un-optimized baseline ensembles. The architecture is presented in Fig. 3.

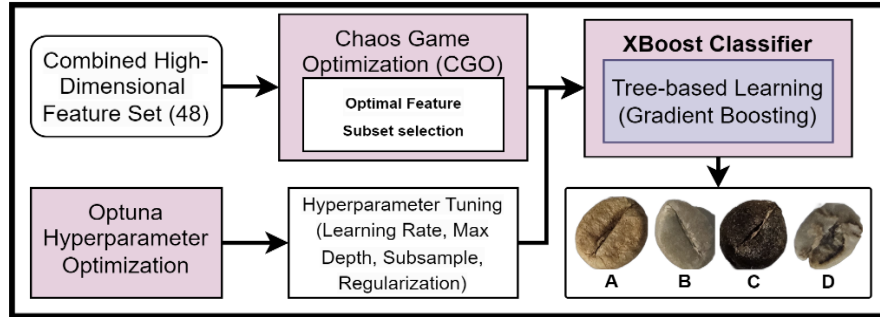


Fig. 3. Schematic overview of the proposed grading framework. The pipeline reduces the initial feature superset using CGO, optimizes the classifier using Optuna, and evaluates the performance using stratified cross-validation.

Feature Selection via Chaos Game Optimization (CGO). As an alternative to utilizing all 48 handcrafted features for training our classifiers, feature selection has been carried out by means of CGO in order to bypass such redundancy since a large number of spectral and textural descriptors inflate computation and hamper generalization. Unlike classical GAs, which converge prematurely to a solution at roughly 89% test accuracy in our experiments, CGO maintains a higher diversity in search space due to

chaotic perturbations. The candidate feature subsets (agents) evolve according to the following as represented in Eq (2).

$$p_i^{(t+1)} = 0.5 (p_i^{(t)} + p_j^{(t)}) + r (\text{best}^{(t)} - p_i^{(t)}) \quad (2)$$

where $p_i^{(t)} \in [0,1]^d$ represents the continuous probability vector of the i -th agent, $p_j^{(t)}$ is a randomly sampled peer, $\text{best}^{(t)}$ is the global best solution, and $r \sim U(0,1)$. The continuous vectors are binarized via a sigmoid-transfer thresholding scheme (values $> 0.5 \rightarrow$ feature selected). This wrapper adopts stratified 5-fold weighted F1 as fitness, that makes the wrapper sensitive to class imbalance.

CGO consistently converged to a subset of 30 features on runs of 80 iterations with a population of 80 agents. Weakly informative Fourier harmonics were eliminated, but high-signal descriptors such as GLCM homogeneity/energy, LBP uniformity, and Gabor magnitude responses were kept in the representative subset. This dimensionality reduction increased the wrapper's cross-validated F1 score from 0.930 to 0.936, indicating that dimensionality pruning rather than merely reducing the model size is the primary factor that directly aids generalization.

Model Optimization: XGBoost with Optuna. For the classification engine, the decision was to go with XGBoost (Extreme Gradient Boosting) instead of LightGBM and Random Forest. While LightGBM had similar inference speeds, XGBoost proved to be much more stable and better regularized on the feature set chosen by CGO and reached top classification metrics in the final evaluation.

The Optuna framework with a Tree-structured Parzen Estimator sampler was utilized to maximize generalization. The optimization objective was to maximize the weighted F1-score on an internal validation split (20%). The hyperparameter search space along with the final values leading to 93.61% accuracy are:

- **Learning Rate (η):** It was Tuned to 0.05 in order to prevent fast convergence to a sub-optimal minimum and at the same time make the gradient descent stable.
- **Tree Depth (max_depth):** Limited to 10. At this depth, the model is able to capture complicated nonlinear texture patterns of coffee beans without the problem of overfitting.
- **Regularization:** The $L1(\alpha)$ and $L2(\lambda)$ penalties were changed to 0.1 and 1.0, respectively. This gave very important regularization to the selected features that had been reduced.
- **Stochastic Subsampling:** The subsample and colsample_bytree parameters were tuned to 0.8 to introduce randomness for further variance reduction.

The proposed approach achieves high precision for automated coffee bean grading while retaining a lightweight computational footprint suitable for edge deployment by integrating the discriminative power of CGO-selected features with the regularized boosting of XGBoost.

3 Results and Discussion

This section details the empirical findings of the proposed automated coffee bean grading system. We conducted experiments to evaluate each stage of the process—segmentation, baseline classification, and the effectiveness of the new meta-heuristic optimization. The complete process is illustrated in Fig. 4.

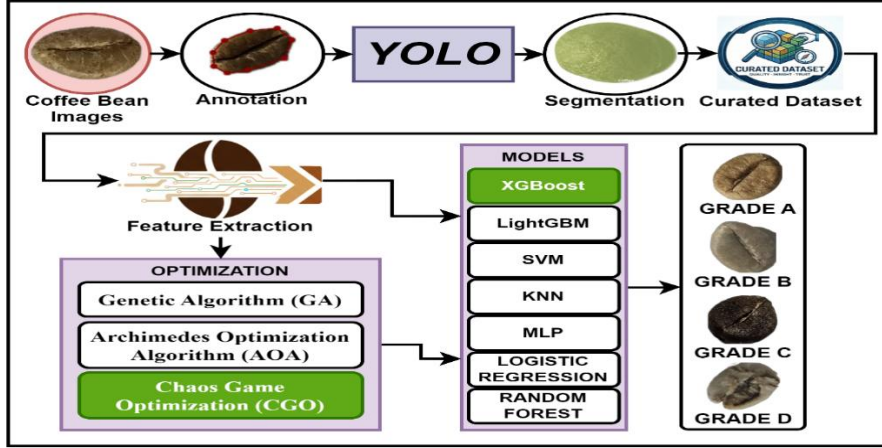


Fig. 4. The experimental workflow.

Ensuring statistical reliability, all models were assessed with a stratified 80:20 train-test division. Accuracy, Precision, Recall, F1-Score, and ROC-AUC were the metrics used to quantify the performance.

3.1 Segmentation Model Performance Analysis

The YOLOv11m-seg model was very effective in the extraction of beans, thus, the results were very reliable. For bounding box detection, it achieved a Precision of 0.998 and a Recall of 0.997 on a test set of 457 images (474 bean instances). Its mAP@50 was 0.995, and mAP@50–95 was 0.969. In the case of mask segmentation, Precision was 0.996 and Recall 0.995, with mAP@50 at 0.989 and mAP@50–95 at 0.872 (see Table 3). There were almost no mistakes. The training went uninterrupted for 100 epochs. The final mAP@50–95 was 0.9757 for training and 0.9688 for validation, which is an indication of a good level of model generalization. The module was able to execute the task very quickly about 12 milliseconds per image on an RTX 4050, thus, real-time, high-throughput sorting can be done without any problem.

Table 3. yolov11m-seg on test set

Metric Type	Precision	Recall	mAP@50	mAP@50–95
Bounding Box	0.998	0.997	0.995	0.969
Mask (Segmentation)	0.996	0.995	0.989	0.872

3.2 Analysis of Individual Feature Sets

To determine the discriminative power of individual visual attributes (see Fig. 5), classifier performance was evaluated on isolated feature subsets. Feature-subset ablations were used to ascertain the contributory value of chromatic, textural, and shape cues. Results (Table 4) confirm a clear hierarchy: Color > Texture > Shape. The LAB descriptors led as the most informative, with 86.84% accuracy and 86.77% F1 with an MLP, thereby confirming spectral shifts, especially in black or sour beans, which provide the most reliable defect signal. Texture features occupied the second tier. Local Binary Pattern captured well surface irregularities, with an F1 score of 80.65%, while Wavelet features supplied multi-resolution detail but introduced additional noise with an F1 of 78.10%. Shape descriptors provided the least information: Fourier features led to a drop in accuracy to 66.19%, which means that global geometry cannot be a reliable factor to differentiate defective beans from healthy ones.

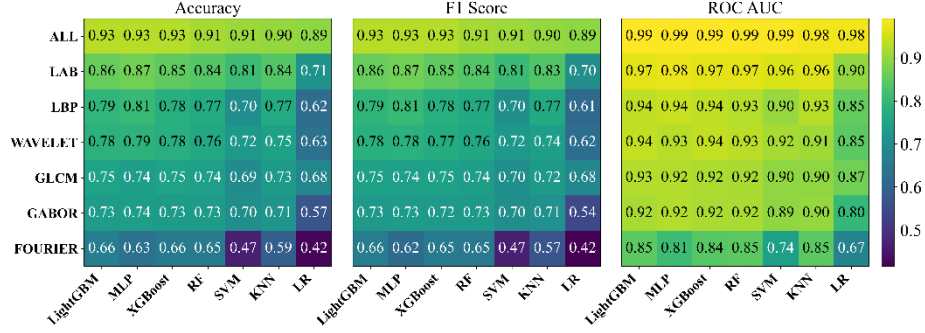


Fig. 5. Shows the performance matrices across three metrics: (a) Accuracy, (b) F1-Score, and (c) ROC-AUC providing analysis of individual features.

The heatmaps indicate the contribution of each grading feature where LAB is consistently the most influential and Fourier are consistently the least.

Table 4. Performance Comparison of Individual Feature Sets

Feature Set	Descriptor Type	Best Model	Test Accuracy	Test F1-Score
LAB	Color	MLP	86.84%	86.77%
LBP	Texture	MLP	80.52%	80.65%
Wavelet	Frequency	LightGBM	78.45%	78.10%
GLCM	Texture	LightGBM	74.97%	74.53%
Gabor	Texture	MLP	73.55%	73.28%
Fourier	Shape	LightGBM	66.19%	65.65%

3.3 Baseline Model Performance (Full Feature Set)

We set a performance benchmark with no dimensionality reduction, using the full 48-dimensional fused feature vector. Among the seven classifiers (see Table 5), LightGBM and MLP produced the best results, each achieving an accuracy of

93.03%, with identical F1-scores of about 93%, which further points to the fact that nonlinear models can make full use of the heterogeneous feature space. Tree-based models, represented by XGBoost and Random Forest, remained competitive but slightly worse. Distance-based methods were clearly at a disadvantage: KNN had a train–test gap in F1-score of around 10%, reflecting overfitting induced by sparsity in the 48-D space. Logistic Regression performed as expected, but was only able to capture the complex decision boundaries in a limited way.

Table 5. Phase 1: Baseline Performance using 'ALL' Features

Model	Test Accuracy	Test F1-Score	ROC-AUC	Overfit Delta (F1)
LightGBM	93.03%	93.02%	0.9905	0.069
MLP Classifier	93.03%	93.01%	0.9914	0.069
XGBoost	92.65%	92.64%	0.9897	0.073
Random Forest	91.35%	91.38%	0.9873	0.068
SVM	90.58%	90.61%	0.9876	0.029
KNN	89.94%	89.97%	0.9785	0.100
Logistic Regression	89.29%	89.26%	0.9780	0.009

3.4 Optimization Strategy and Dimensionality Reduction

For the comparison of the influence of each optimization strategy on all classifiers, the heatmaps of Accuracy and F1-Score performances were generated (Fig. 6).

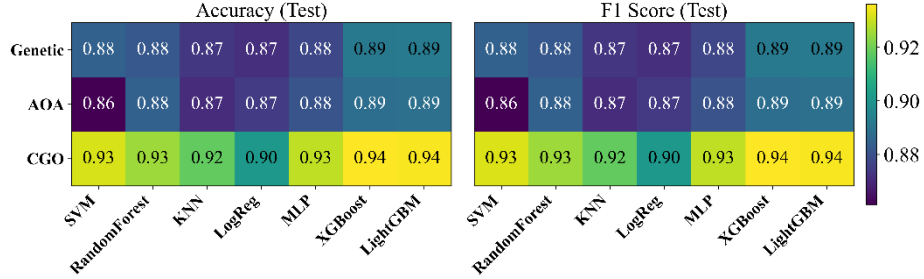


Fig. 6. Heatmap of Testing Accuracy and F1-Score: Optimization Algorithms vs. Classifiers.

Heatmaps interpretation. The heatmaps are structured by having Optimizers as rows and Classifiers as columns. The intensity of color, lighter being higher performance, will show three clear patterns:

- Dominance of CGO: Bottom Row. Notice that for both metrics, the row corresponding to CGO is brightest, indicating that the CGO-selected subset is universally more discriminative, regardless of the choice of downstream classifier.
- Classifier Sensitivity.
 - Tree-Based Models (XGBoost, LightGBM): These which appear in the rightmost columns. They are very robust even to weaker feature subsets,

- GA/AOA, retaining about 89% accuracy. Under CGO, they peak at 93.6%, showing that they fully utilize the richer feature manifold identified by CGO.
- Distance- and Linear-Based Models (SVM, KNN, Logistic Regression): These represent the most significant gain with CGO. For example, SVM goes from approximately 86% AOA to 93.16% CGO, evidence that CGO finds a feature space where classes are more linearly separable, benefiting models that do not perform tree-based partitioning.
- Stability Across Models. The CGO row has a low variance across classifiers, while the results for both GA and AOA present incoherent performances, with significantly worse outcomes for Logistic Regression and SVM. This points to the robustness of the subset chosen by CGO.

The handcrafted feature set is integrated with Chaos Game Optimisation (CGO)–based feature selection and Optuna-tuned XGBoost under cross-validation to form a compact and empirically reliable classification pipeline. The proposed framework achieves a test accuracy of 93.61%, surpassing the strongest baseline (93.03%) by 0.58%. Although numerically modest, this gain is practically meaningful: CGO compresses the original 48-dimensional feature space to 30 features without degrading performance, reducing model complexity and inference latency and enabling real-time deployment on Raspberry Pi and Jetson Nano platforms. The limited train–test gap (100% $\rightarrow$ 93.6%) indicates improved generalization and reduced sensitivity to optimizer-induced overfitting. Unlike end-to-end deep models, the proposed design preserves feature-level interpretability while maintaining competitive accuracy under constrained computational budgets. The current evaluation is restricted to single-bean Robusta images acquired under controlled conditions; therefore, extending validation to multi-bean scenes, additional coffee varieties, uncontrolled acquisition environments, and benchmarking against CNN and Transformer architectures is essential.

4 Conclusion

This work presents a complete computer vision pipeline for grading pre-roast Robusta coffee beans. A YOLOv11m-seg–based segmentation stage reliably localizes individual beans and extracts standardized regions of interest. Through ablation analysis, CIELAB color features and Local Binary Patterns (LBP) are shown to contribute most significantly to classification performance, while geometric features offer negligible discriminative power. The key technical contribution is the application of Chaos Game Optimization (CGO) for feature selection, which demonstrates superior convergence behavior compared to Genetic Algorithms (GA) and the Arithmetic Optimization Algorithm (AOA). Combined with an Optuna-tuned XGBoost classifier, the proposed system achieves a test accuracy of 93.61% with a computational footprint suitable for deployment on edge devices such as Raspberry Pi and NVIDIA Jetson Nano.

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